

REQUIREMENT ANALYSIS

Electricity Consumption Analysis

Customer Journey Map

- **Step 1: Identify the Problem**
The user identifies the need to analyze electricity consumption data across different states and regions to understand usage patterns and support better energy planning.
- **Step 2: Collect Data**
Electricity consumption data is collected from a reliable dataset containing information such as states, regions, dates, latitude, longitude, and electricity usage.
- **Step 3: Store and Analyze Data**
The dataset is imported into MySQL. SQL queries are executed to calculate total usage, average usage, minimum usage, state-wise consumption, and region-wise consumption.
- **Step 4: Create Visualizations**
Interactive charts, maps, and graphs are developed using Tableau Desktop to visualize electricity consumption patterns.
- **Step 5: Generate Insights**
Business insights are generated by identifying high electricity-consuming states, comparing regional usage, analyzing monthly trends, and recognizing consumption patterns.
- **Step 6: Publish Dashboard**
The Tableau dashboard is published on Tableau Public and embedded into a Flask web application for easy access through a web browser.
- **Step 7: Monitor and Improve**
Users monitor electricity consumption trends using the interactive dashboard and use the insights to support better energy management and planning.

Solution Requirements

Functional Requirements

- **FR-1 Dataset Import**
The system shall allow importing the electricity consumption dataset into the MySQL database for analysis.
- **FR-2 Data Storage**
The system shall securely store electricity consumption records in MySQL for efficient querying and management.

➤ **FR-3 SQL Analysis**

The system shall execute SQL queries to calculate total usage, average usage, minimum usage, state-wise usage, and region-wise usage.

➤ **FR-4 Visualization Creation**

The system shall generate interactive charts, graphs, maps, and trend lines using Tableau Desktop.

➤ **FR-5 Dashboard Development**

The system shall provide an interactive dashboard displaying electricity consumption across states, regions, and months.

➤ **FR-6 Story Creation**

The system shall create Tableau Story pages that explain the project overview, analysis, business insights, and conclusions.

➤ **FR-7 Dashboard Publishing**

The system shall publish the dashboard to Tableau Public and integrate it with a Flask web application.

Non-Functional Requirements

➤ **NFR-1 Usability**

The dashboard should be simple, intuitive, and easy for users to understand.

➤ **NFR-2 Reliability**

The system should provide accurate and consistent analysis based on the dataset.

➤ **NFR-3 Performance**

The dashboard should load quickly and respond efficiently during user interactions.

➤ **NFR-4 Availability**

The published dashboard should be accessible online through Tableau Public and the Flask application.

➤ **NFR-5 Scalability**

The system should support larger electricity consumption datasets in future implementations.

➤ **NFR-6 Maintainability**

The dashboard, SQL queries, and Flask application should be easy to update and maintain when new data becomes available.

Data Flow Diagram

